Microeconomics I

Class 7 — The Slutsky equation and the money value of a price change

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Plan for today

Warm-up; a mug, three prices

The Slutsky equation: substitution and wealth effects

Giffen goods, revisited

In pairs: gross and net cross effects

Break

The Slutsky matrix; the map of x, v, h, e

Compensating and equivalent variation (CV, EV)

In pairs: a cheaper good

Deadweight loss; summary

Practicalities

- Problem set 2: due at the start of class, Wed 21 Oct — on paper, one copy per group
- Office hours: Wednesdays 14:00–15:00, Room 445

  up to 20 minutes per student; those who book by e-mail have priority
- Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu

Warm-up: always true, or not?

Last class. True for every consumer? One minute with your neighbour.

p_1 rises $\Rightarrow$
$h_1(p, u)$ does not rise

p_1 rises $\Rightarrow$
$x_1(p, w)$ does not rise

$e(p, u)$ is homogeneous of degree 0 in p

$e\Big(p, v(p, w)\Big) = w$

Class 6, slides 10, 13, 14; class 5, slide 22. MWG Prop. 3.E.2, 3.E.4; (3.E.1).

A mug, three groups: whose prices should be equal?

Simon Fraser University: 77 students, a coffee mug, three groups at random

Sellers are given a mug: the lowest price at which they sell it

Buyers: the highest price they pay for one

Choosers: a mug or cash: the cash that is worth as much as the mug

Which two medians should standard theory make equal?

buyers and choosers · choosers and sellers · buyers and sellers

A published experiment; the source in class.

A mug, three groups: what they said

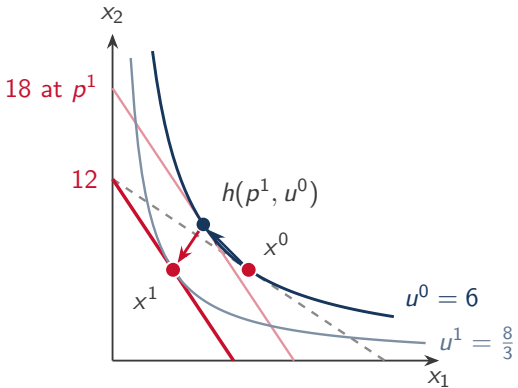
Revealed in class

Before class: which two groups face exactly the same choice?

The Slutsky equation

MWG 3.G

Good 1 dearer: x_1 falls $3 \to 2$ along u^0, then $2 \to \frac{4}{3}$ from lost wealth



x_1: $3 \to 2 \to \frac{4}{3}$

substitution -1, wealth $-\frac{2}{3}$

x_2: $2 \to 3 \to 2$

substitution $+1$, wealth -1

18 euros at p^1 keep her on u^0

Substitution: along u^0, at the new prices.

Wealth: the parallel shift from 18 euros down to her 12.

MWG 3.G, pp. 71–72; class 6, slide 14. $u = x_1 x_2$, $w = 12$, $p^0 = (2, 3) \to p^1 = (4.5, 3)$.

Slutsky equation: compensated slope = market slope + wealth term

$$\frac{\partial h_\ell(p,u)}{\partial p_k} = \frac{\partial x_\ell(p,w)}{\partial p_k} + \frac{\partial x_\ell(p,w)}{\partial w}\, x_k(p,w) \quad \text{at } u = v(p,w)$$

why: differentiate $h_\ell(p,u) = x_\ell\big(p, e(p,u)\big)$ in p_k; $\partial e/\partial p_k = h_k = x_k$ (Shephard)

$$\frac{\partial x_\ell}{\partial p_\ell} = \underbrace{\frac{\partial h_\ell}{\partial p_\ell}}_{\leq 0\text{: substitution}} \underbrace{- x_\ell \frac{\partial x_\ell}{\partial w}}_{\text{wealth}}$$

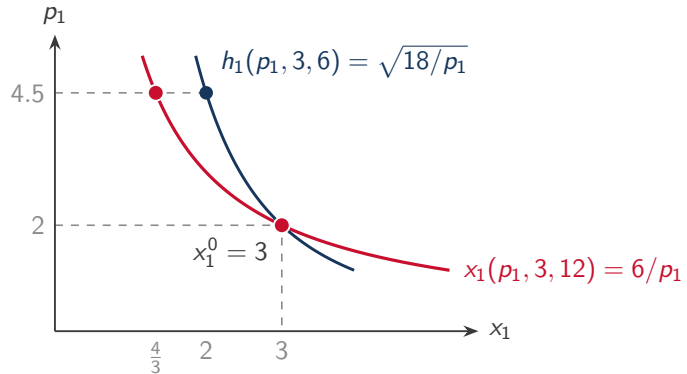
$x_1 x_2$ at $p = (2,3)$, $w = 12$: $-\frac{3}{2} = -\frac{3}{4} - 3 \cdot \frac{1}{4}$

A normal good falls more in market demand than in compensated demand.

MWG Prop. 3.G.3, p. 71, (3.G.3)–(3.G.4): u continuous, $\succsim$ locally non-satiated and strictly convex.

Market demand falls faster than compensated: good 1 is normal

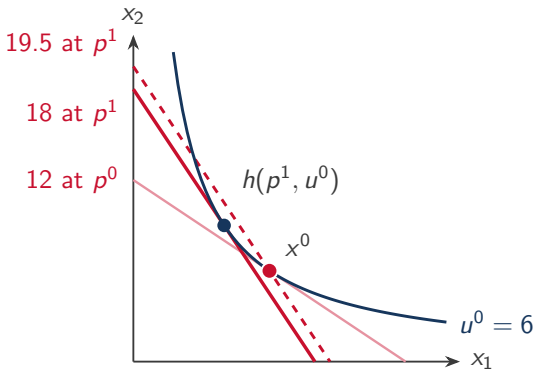
$u = x_1 x_2$, $p_2 = 3$, $w = 12$, $u^0 = 6$: both curves pass through $x_1 = 3$ at $p_1 = 2$



At $p_1 = 4.5$: compensated demand 2, market demand $\frac{4}{3}$; the gap is the wealth effect.

MWG Figure 3.G.1(a), p. 71; text p. 72.

Two ways to compensate: keep u^0 (6 euros) or x^0 affordable (7.5)



Hicks: $e(p^1, u^0) - w = 18 - 12 = 6$

back on her old curve

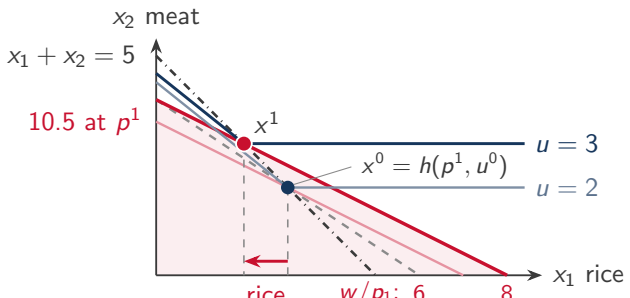
Slutsky: $p^1 \cdot x^0 - w = 19.5 - 12 = 7.5$

her old bundle affordable again

Hicks $\leq$ Slutsky, equal to first order: so the compensated slopes of slide 9 carry Slutsky's name.

MWG 3.G, pp. 72–73, Figure 3.G.2; class 6, slide 22.

Giffen needs inferior: class 5's rice, $3 \to 3 \to 2$



rice p_1: $2 \to 1.5$

x_1: $3 \to 3 \to 2$

substitution 0 (the kink on the food line $x_1 + x_2 = 5$), wealth -1

at $p_1 = 2$:

$$\frac{\partial x_1}{\partial p_1} = 0 - 3 \cdot (-1) = +3$$

$\partial h_1/\partial p_1 = 0$, $x_1 = 3$, $\partial x_1/\partial w = -1$

Substitution is never positive, so $\partial x_\ell/\partial p_\ell > 0$ needs $\partial x_\ell/\partial w < 0$: Giffen needs inferior.

Class 5, slides 19 and 22: $u = \min\{x_2,\ x_2 - 4(5 - x_1 - x_2)\}$, $w = 12$, $p_2 = 3$. MWG Prop. 3.G.3, p. 71.

In pairs: net cross effect = gross + wealth term

Class 6, slide 21, at its start: $p = (2, 3)$, $w = 12$. x_2 from the budget. $\sqrt{x_1} + \sqrt{x_2}$: the CES with $\rho = \frac{1}{2}$.

u	$x_1(p, w)$	gross $\frac{\partial x_1}{\partial p_2}$	wealth term $x_2 \frac{\partial x_1}{\partial w}$	net $\frac{\partial h_1}{\partial p_2}$
$\min\{x_1, x_2\}$	$\frac{w}{p_1 + p_2}$			
$x_1 x_2$	$\frac{w}{2p_1}$			
$\sqrt{x_1} + \sqrt{x_2}$	$\frac{w\, p_2}{p_1(p_1 + p_2)}$			

Four minutes.

Answers revealed in class. Try it before class.

(optional) The Slutsky equation in elasticities

$\varepsilon_{11} = (\partial x_1/\partial p_1)\, p_1/x_1$, $\varepsilon_{1w} = (\partial x_1/\partial w)\, w/x_1$, $\varepsilon^h_{11} = (\partial h_1/\partial p_1)\, p_1/h_1$;
budget share $b_1 = p_1 x_1/w$.

(1) Show: $\varepsilon_{11} = \varepsilon^h_{11} - b_1\, \varepsilon_{1w}$.

(2) $\varepsilon^h_{11} = -0.2$. For which ε_{1w} is good 1 Giffen if $b_1 = 0.4$? If $b_1 = 0.02$?

At home if not done in class; the answers are in the slides released after class.

MWG (3.G.3), p. 71; elasticities as in MWG 2.E, p. 27; budget share b_ℓ, p. 28. Invented numbers in (2).

Break

Ten minutes

The Slutsky matrix: class 6's $D_p h$, built from observable demand

$$S(p, w): \quad s_{\ell k} = \frac{\partial x_\ell}{\partial p_k} + \frac{\partial x_\ell}{\partial w} x_k$$

only Walrasian demand: observable in principle

$$S = \begin{pmatrix} -\frac{3}{2} + 3 \cdot \frac{1}{4} & 0 + 2 \cdot \frac{1}{4} \\ 0 + 3 \cdot \frac{1}{6} & -\frac{2}{3} + 2 \cdot \frac{1}{6} \end{pmatrix} = \begin{pmatrix} -\frac{3}{4} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{3} \end{pmatrix}$$

$= D_p h(p, 6)$: class 6, slide 20

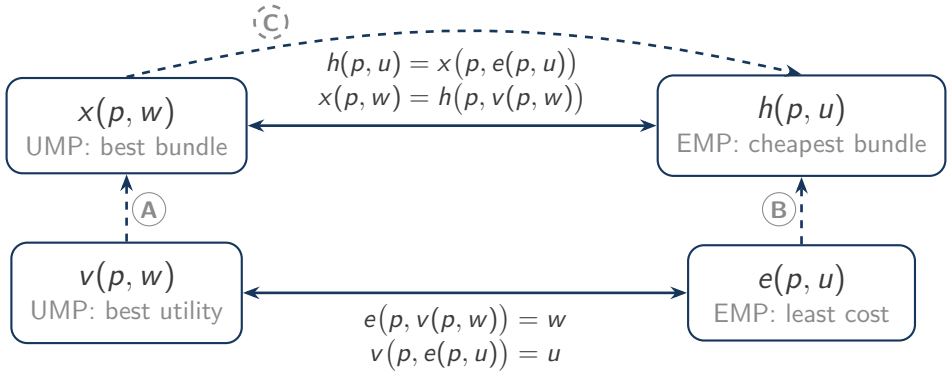
negative semidefinite symmetric $S\,p = 0$

Demand whose S is not symmetric or not negative semidefinite comes from no rational preferences.

MWG Prop. 3.G.2, pp. 69–70; 3.G, pp. 72–73: $S(p, w) = D_p h(p, u)$ (Prop. 3.G.3).

In pairs: the map of x, v, h, e — name the three derivative arrows

In pairs: name A, B, C — formula and name.



MWG Figure 3.G.3, p. 75 (a similar figure: Deaton and Muellbauer 1980); Props. 3.G.1, 3.G.3, 3.G.4.

Money measures of welfare

MWG 3.I

Price the change in euros: EV at the old prices, CV at the new

money metric: $e\big(\bar{p}, v(p, w)\big)$

utility in euros at fixed prices $\bar{p}$

equivalent variation: $\mathrm{EV} = e(p^0, u^1) - w$

at the old prices, the change in wealth as good as the price change: $v(p^0, w + \mathrm{EV}) = u^1$

compensating variation: $\mathrm{CV} = w - e(p^1, u^0)$

at the new prices, what can be taken after the change to restore u^0: $v(p^1, w - \mathrm{CV}) = u^0$

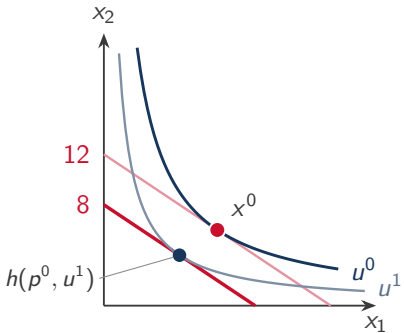
$u^0 = v(p^0, w), \quad u^1 = v(p^1, w)$

EV and CV: both negative when she loses

Both rank p^0 and p^1 correctly; their euros differ.

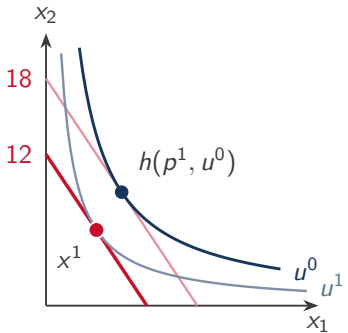
MWG 3.I, pp. 81–82: Figure 3.I.1; (3.I.1), (3.I.2), after Hicks (1939).

Good 1 to 4.5: EV $= -4$ euros, CV $= -6$ euros



EV: old prices

$e(p^0, u^1) - w = 8 - 12 = -4$



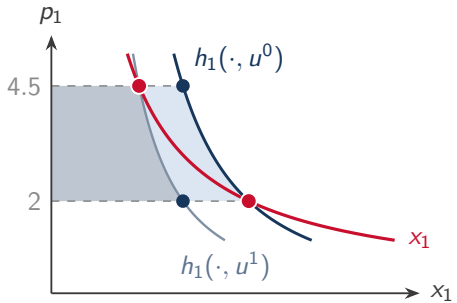
CV: new prices

$w - e(p^1, u^0) = 12 - 18 = -6$

Old prices value her loss at 4 euros, new prices at 6: EV $>$ CV, good 1 is normal.

MWG Figure 3.I.2, p. 82; $u = x_1 x_2$, $w = 12$, $p^0 = (2, 3)$, $p^1 = (4.5, 3)$.

$-\mathbf{CV} = 6$ and $-\mathbf{EV} = 4$ are areas left of Hicksian demand



$-$CV: both shades $-$EV: dark

$$-\text{CV} = \int_2^{4.5} h_1(p_1, 3, u^0)\, dp_1 = 6$$

$$-\text{EV} = \int_2^{4.5} h_1(p_1, 3, u^1)\, dp_1 = 4$$

$$\int_2^{4.5} x_1(p_1, 3, 12)\, dp_1 = 6 \ln 2.25 \approx 4.87$$

market demand: in between

Shephard turns a change in e into an area under compensated demand.

MWG 3.I, pp. 83 and 89: (3.I.3), (3.I.4), (3.I.8); Figure 3.I.3.

Revealed in class

Before class: which group's price is a CV of getting a mug, which an EV?

In pairs: good 1 falls to 0.5: what is the gain worth?

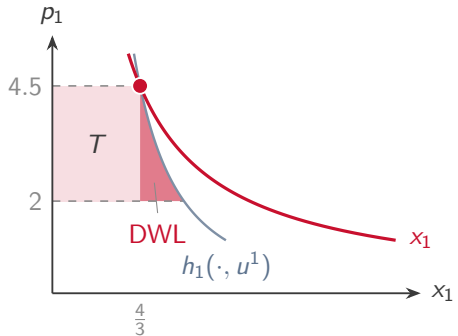
$$u = x_1 x_2, \quad w = 12, \quad p^0 = (2, 3) \to p^1 = (0.5, 3).$$

question	answer
new bundle x^1 and utility u^1	
$h(p^1, u^0)$: split x_1 into substitution and wealth	
$\text{CV} = w - e(p^1, u^0)$	
$\text{EV} = e(p^0, u^1) - w$	

Five minutes.

Answers revealed in class. Try it before class.

Tax good 1 by 2.5: it raises 3.33, costs her 4: deadweight loss 0.67



tax $t = 2.5$ per unit: p_1: $2 \to 4.5$

revenue $T = t\, x_1(p^1, w) = 2.5 \cdot \frac{4}{3} \approx 3.33$

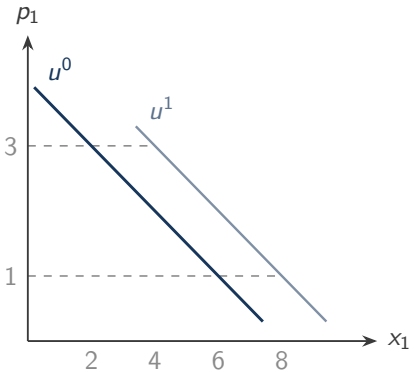
her loss $-\text{EV} = 4$

deadweight loss $= 4 - 3.33 \approx 0.67$

The tax takes more from her than it brings in: a triangle left of Hicksian demand.

MWG Example 3.I.1, pp. 84–85, (3.I.5), Figure 3.I.4(a); own numbers.

(optional) CV and EV from two compensated demand curves



p_1: $1 \to 3$; p_2, w fixed.

$h_1(p_1, u^0) = 8 - 2p_1$, $h_1(p_1, u^1) = 10 - 2p_1$

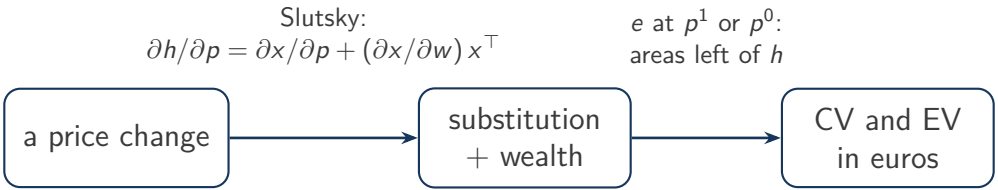
(1) CV and EV?

(2) Normal or inferior?
Is EV $>$ CV, as on slide 20?

At home if not done in class; the answers are in the slides released after class.

MWG 3.I, p. 83: (3.I.3), (3.I.4), Figure 3.I.3. Own numbers.

Today in one line



Wednesday: Uncertainty I: lotteries, the independence axiom, the expected utility theorem and the uniqueness of the vNM utility function.

Money lotteries, risk aversion, certainty equivalent, risk premium.

Problem set 2: due at the start of class, Wednesday 21 October — on paper, one copy per group.

Read MWG 6.B–6.C (or Muñoz-Garcia ch. 5) for Wednesday 21 October.